EchoSense

LSTM Autoencoder for Monitoring Ecosystem Degradation in Vellore

LSTM Autoencoder · Three-Signal Fusion · Silence Score™

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Presentation Outline

OVERVIEW

01 **Problem & Motivation**
Why Vellore needs season-aware monitoring

02 **Vellore Observatory**
Four local ecosystems and a fixed study area

03 **LSTM Autoencoder**
Five-step NDVI sequence reconstruction

04 **Three-Signal Fusion**
Reconstruction, Z-score and seasonal deviation

05 **Silence Score**
Interpretable risk scale from 0 to 100

06 **Literature Review**
Foundations of NDVI and sequence anomaly detection

07 **Coverage & Diagnostics**
Dataset scope and current inference outputs

08 **Vellore Sector Profiles**
River basin, forest, lake and agro-corridor

09 **Dashboard & Deployment**
FastAPI, 3D globe, analytics and weather context

10 **Conclusion & Future Work**
Current contributions and next steps

Background & Motivation

CHAPTER 1

Vellore's Semi-Arid Context

Dry summers and monsoon-driven vegetation cycles create different stress patterns across forests, wetlands, riverbanks and farms.

NDVI — A Vegetation Health Signal

NDVI measures vegetation greenness from red and near-infrared reflectance. EchoSense analyses its monthly time-series pattern.

Monsoon-Driven Seasonality

Vellore shifts between a dry summer, southwest monsoon, northeast monsoon and winter-dry period. A seasonal dip is not automatically degradation.

Why Local Context Matters

One fixed NDVI cut-off can confuse normal summer decline with ecosystem stress. EchoSense compares sequence behaviour with a local rolling baseline.

Vellore Observatory

0.6 – 0.9	Dense Healthy Forest
0.4 – 0.6	Moderate Vegetation
0.2 – 0.4	Sparse / Stressed
0.0 – 0.2	Bare Soil / Desert
< 0.0	Water / Built-up

Problem Statement

CHALLENGE

Three Monitoring Gaps EchoSense Addresses

01

Static NDVI Thresholds

A single cut-off ignores differences between seasons and ecosystems. The same NDVI value can have different meaning in a dry month and after monsoon rainfall.

02

Point-in-Time Analysis

A single reading hides the direction and pace of change. Trend, slope and reconstruction behaviour provide more context than an isolated NDVI value.

03

Fragmented Local Monitoring

NDVI analysis, weather context, sector maps and reports are often viewed separately. EchoSense brings these views together for four Vellore sectors.

Project Objectives

GOALS

LSTM Autoencoder

Reconstruct five-step NDVI windows using three features: NDVI, slope and rolling mean. Reconstruction error signals departures from learned patterns.

Environmental Context

Fetch current Vellore weather through Open-Meteo, cache responses for 10 minutes and use a climatological fallback when offline.

Vellore Sector Monitoring

Configure four mapped ecosystems: Palar Basin, Jawadhu Foothills, Otteri Lake and Katpadi Agro-Corridor.

Full-Stack Observatory

Deliver FastAPI inference, a geospatial dashboard, CSV upload, deep analytics and downloadable sector reports.

Three-Signal Fusion

Combine reconstruction error (50%), Z-score deviation (30%) and seasonal deviation (20%) into the Silence Score.

System Architecture Overview

ARCHITECTURE

DATA + WEATHER CONTEXT	Monthly NDVI series and optional CSV uploads; live weather is displayed as contextual telemetry.
PREPROCESSING	Per-series MinMax scaling · NDVI difference (slope) · 3-step rolling mean · 5-step sequence windows
LSTM AUTOENCODER	PyTorch encoder (3 → 64) and decoder (64 → 3) · One LSTM layer each · Reconstruction MSE
STATISTICAL SIGNALS	Z-score of reconstruction error · Absolute NDVI deviation from the 3-step rolling mean
FUSION ENGINE	0.50 × MSE + 0.30 × Z-score + 0.20 × seasonal deviation · Silence Score scaled to 0–100
CLASSIFICATION	Healthy (<30) · Stress (30–70) · Critical (>70) · Separate anomaly flag: mean + 1.5σ

Data Preprocessing & Feature Engineering

PREPROCESSING

Step 1 · Feature Scaling

MinMaxScaler scales NDVI, slope and rolling mean to [0, 1]. The scaler is fitted to the series being processed.

Step 2 · NDVI Slope

The first difference (NDVI.diff) captures the direction and rate of vegetation change between consecutive months.

Step 3 · Rolling Baseline

A 3-step rolling mean provides local seasonal context. Initial missing values are back-filled before inference.

Step 4 · Sequence Windowing

Overlapping windows of 5 time steps are created from the three scaled features and passed to the LSTM Autoencoder.

LSTM Autoencoder Architecture

DEEP LEARNING

Input	Tensor (batch, 5, 3): NDVI + slope + rolling mean
Encoder	Single-layer LSTM · input features = 3 · hidden size = 64
Bottleneck	Final 64-unit encoder state is repeated across the 5 time steps
Decoder	LSTM decoder · input size = 64 · output features = 3
Output	Reconstructed tensor (batch, 5, 3) · compared with input using MSE
Training	Inference loads trained weights from backend/echosense_model.pth

How Inference Works

1. Build 5-step windows from the scaled NDVI features.
2. Reconstruct each window with the trained LSTM Autoencoder.
3. Pass reconstruction error to the Z-score and seasonal fusion stage.

Three Signals Behind the Silence Score

MODEL SIGNALS

Why Combine Three Signals?

The current backend combines temporal reconstruction with statistical deviation and a local NDVI baseline. Weather is shown alongside inference, but is not an LSTM input feature.

1 · Reconstruction Error

Mean squared difference between the input window and its LSTM reconstruction. A larger error means the sequence is less familiar to the model.

2 · Z-Score Deviation

Standardizes reconstruction error against the processed series mean and standard deviation, highlighting unusually high error values.

3 · Seasonal Deviation

Absolute difference between observed NDVI and its 3-step rolling mean adds local trend context to the anomaly score.

Implementation Boundary

Isolation Forest is not called by the current inference service. It remains a possible future comparator, not part of the deployed fusion.

The Fusion Engine & Silence Score

CORE INNOVATION

Three weighted signals, one interpretable output

50% LSTM MSE + 30% Z-Score + 20% Seasonal

The fused value is min–max scaled per processed series to create the Silence Score (0–100).

LSTM Reconstruction Error (weight: 50%)

The main temporal signal. MSE measures how closely each five-step window is reconstructed by the trained model.

Z-Score Deviation (weight: 30%)

Places reconstruction error in the context of the series mean and spread, highlighting unusually high error windows.

Seasonal Deviation (weight: 20%)

Measures the gap between NDVI and its three-step rolling mean. The final score is scaled to 0–100; the anomaly flag uses mean + 1.5σ.

Classification Logic & Health States

CLASSIFICATION

	Interpretation:	Recommended Action:
Score: < 30 **HEALTHY**	NDVI behaviour remains close to the expected pattern for the processed series.	Continue routine monitoring. Review the seasonal context and current weather telemetry.
Score: 30 – 70 **STRESS**	The Silence Score indicates a noticeable departure from the local NDVI pattern.	Review the affected sector, check rainfall and soil moisture, and increase monitoring frequency.
Score: > 70 **CRITICAL**	A strong departure from the reconstructed sequence and local rolling baseline is indicated.	Prioritise field review and local intervention planning for the affected ecosystem.

Literature Review

CHAPTER 2

Rouse et al. (1974) + Tucker (1979)

NDVI is established as a proxy for vegetation health and green biomass; it remains the core input signal in EchoSense.

Myneni et al. (1997)

Global NDVI series show strong seasonal periodicity. This motivates explicit local baselines when interpreting vegetation changes.

Hochreiter & Schmidhuber (1997)

LSTMs retain sequence information over time, making them useful for modelling recurring and non-linear temporal patterns.

Malhotra et al. (2015)

LSTM reconstruction error can be used as an anomaly signal when new sequences differ from learned patterns.

Hundman et al. (2018) — NASA

NASA telemetry work highlighted the importance of threshold selection for reconstruction-based anomaly detection.

Liu et al. introduced a tree-based anomaly detector. It is a relevant comparator, but not used in the current EchoSense inference code.

Introduced Isolation Forest: isolate anomalies via random tree splits. Anomalies require fewer splits → shorter path → higher anomaly score. Computationally efficient, handles high-dimensional feature spaces.

CCDC models seasonal vegetation cycles and structural change, providing a remote-sensing comparison for NDVI monitoring.

The review surveys deep and shallow anomaly detection methods and motivates transparent score-level combination of signals.

Ruff et al. (2021) — Deep AD Review

Categorized hybrid anomaly detection fusion strategies: score-level, feature-level, decision-level. EchoSense employs SCORE-LEVEL FUSION — maximizing interpretability by keeping each component's contribution transparent and auditable.

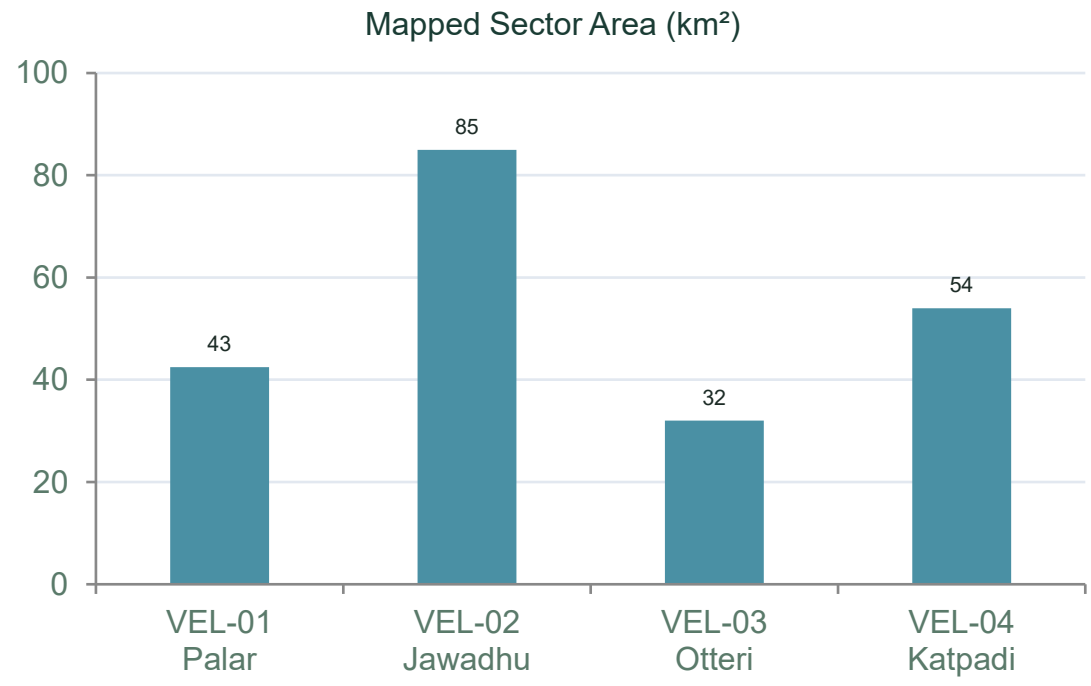
EchoSense vs. Existing Approaches

COMPARISON

Aspect	Static Threshold	Rolling Z-Score	LSTM Autoencoder	EchoSense
Detection Signal	Fixed NDVI cut-off	Statistical deviation	Reconstruction MSE	**Three-signal fusion**
Temporal Context	None	Rolling window	5-step sequence	**5-step + rolling baseline**
Seasonality	Not explicit	Partial	Learned from sequence	**Explicit 3-step baseline**
Adaptability	Fixed rule	Window-dependent	Trained weights	**Per-series scaling + threshold**
Feature Input	Single NDVI value	NDVI + statistics	3-feature windows	**NDVI, slope, rolling mean**
Interpretability	Simple cut-off	Z-score	Reconstruction gap	**Silence Score (0–100)**
Operational Flow	Manual review	Batch analysis	Model inference	**FastAPI + dashboard**

Vellore Monitoring Footprint & Model Diagnostics

DATA + DIAGNOSTICS



0.1082
Mean reconstruction MSE

0.3289
RMSE (scaled features)

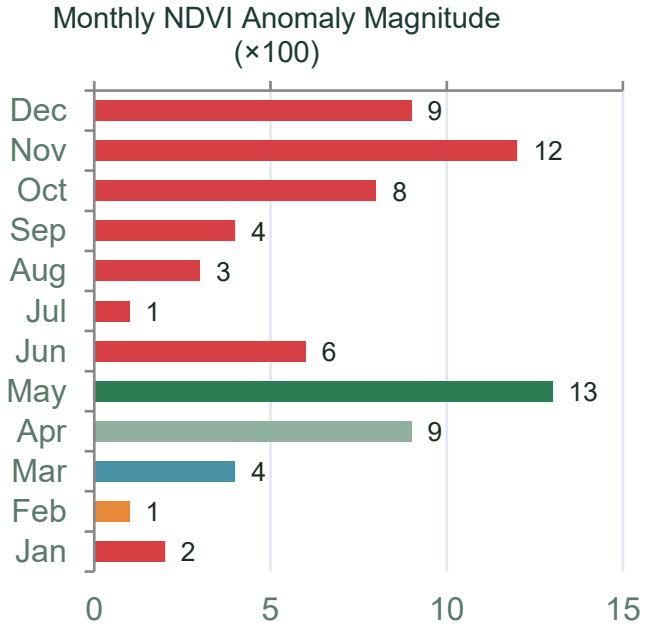
9.3%
Anomaly ratio

48
Monthly NDVI points

Diagnostics shown for the bundled base NDVI series; no labelled hold-out set is supplied for Precision, Recall, F1 or ROC-AUC.

Vellore's Seasonal NDVI Context

LOCAL CLIMATE PROFILE



Summer Stress Window

April–May reach 36.5–38.9°C in the reference profile, alongside NDVI anomalies of −0.09 to −0.13.

Monsoon Greening

October–November rainfall rises to 195–215 mm, with positive NDVI anomalies of +0.08 to +0.12.

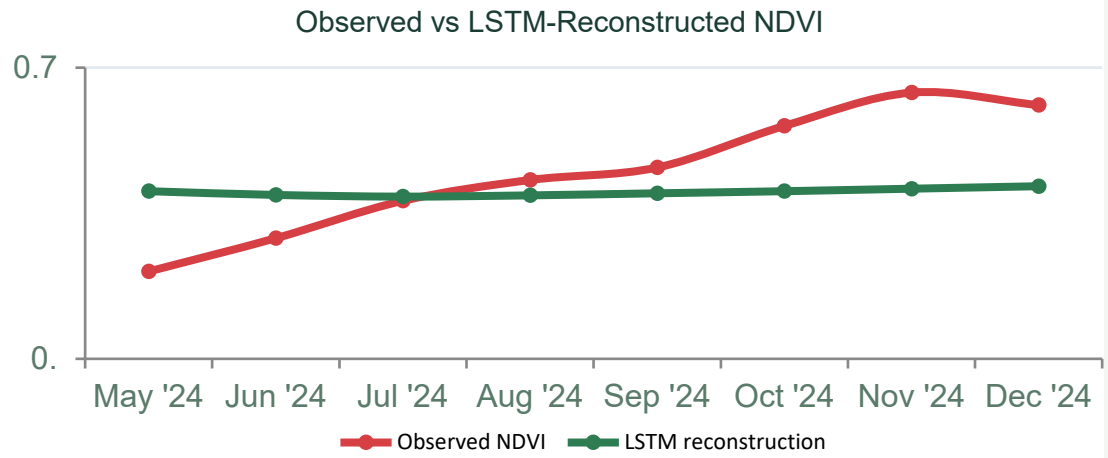
How EchoSense Uses Context

The rolling baseline helps interpret seasonal NDVI change. Weather telemetry is displayed as context, not fed into the LSTM.

Bars show anomaly magnitude; Mar–Jun are negative and Jul–Feb positive in the bundled profile—not live satellite data.

Example Inference — Vellore Base NDVI Series

MODEL OUTPUT



Base Series Window

May–Dec 2024

Eight monthly points from the bundled NDVI base series are shown in this example run.

LSTM Reconstruction

5-step windows

The model reconstructs the expected NDVI pattern from NDVI, slope and rolling-mean inputs.

Stress Band Entry

Nov 2024 · Dec 2024

Silence Scores of 37.1 and 46.3 fall within the Stress band in this base-series inference.

Dynamic Anomaly Flag

Mean + 1.5σ

The statistical anomaly flag is calculated separately from the three display severity bands.

Scope Note

Illustrative output

Base-series inference only; not a field-validated forecast.

Vellore's Four Monitoring Sectors

OBSERVATORY SCOPE

213.5 km² mapped across four distinct local ecosystems



VEL-01 · Palar River Basin

42.5 km² | Riparian scrub and riverbed aquifer. Key concern: pre-monsoon groundwater and drybed moisture deficit.



VEL-02 · Jawadhu Foothills

85.0 km² | Dry deciduous and scrub forest. Monitor canopy moisture during a delayed Northeast Monsoon.



VEL-03 · Otteri Lake Catchment

32.0 km² | Wetland and semi-arid scrub. Sensitive to runoff, tank storage and seasonal desiccation.



VEL-04 · Katpadi Agro-Corridor

54.0 km² | Irrigated agriculture and tank network. Seasonal crop cycling and summer fallow are key context.

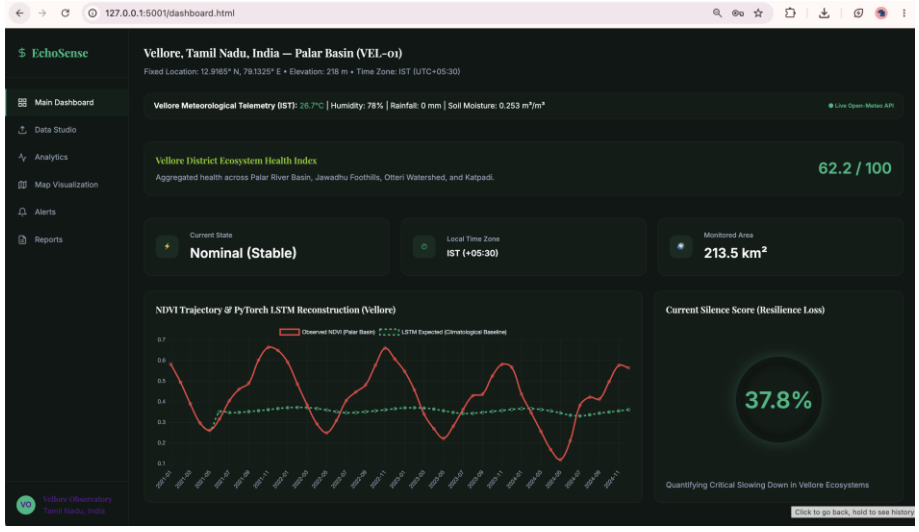
Vellore Observatory — Dashboard & Deployment

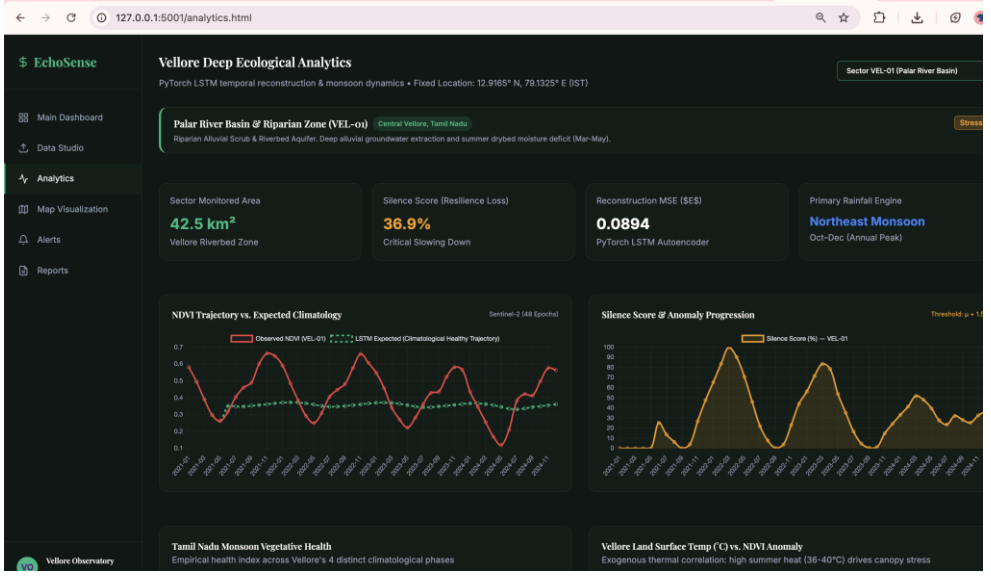
DEPLOYMENT

Geospatial 3D Globe + 2D Map	Globe.gl and Leaflet views display four Vellore sectors with boundaries, ecosystem details and color-coded condition labels.
Chart.js Deep Analytics	Compare observed and LSTM-reconstructed NDVI, track Silence Score changes, and view monsoon and temperature/rainfall context.
Open-Meteo + CSV Live Context & Upload	Weather telemetry is cached for 10 minutes with a climatological fallback. CSV upload runs NDVI inference; reports include sector summaries and CSV export.

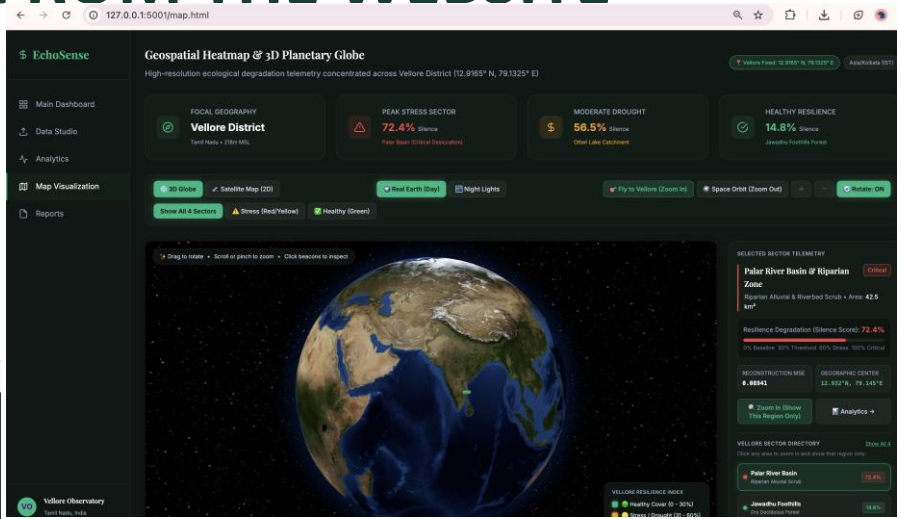
Tech Stack: FastAPI + Uvicorn · PyTorch · pandas / scikit-learn · Leaflet.js + Globe.gl · Chart.js · Vanilla JS

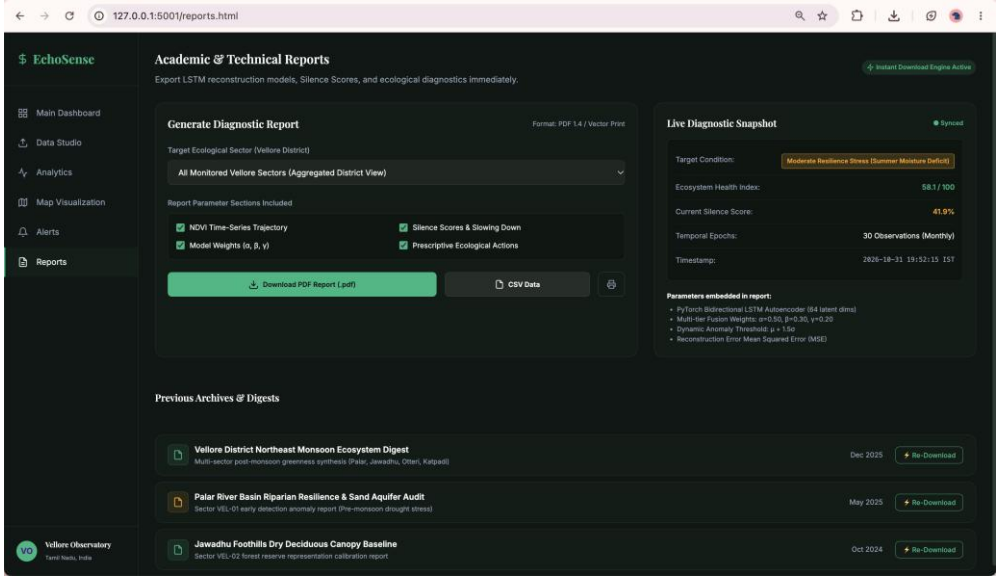
SOME OF THE FEATURES FROM THE WEBSITE





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Real-World Applications & Impact

SIGNIFICANCE

Agro-Ecological Planning

Katpadi sector monitoring can support crop-cycle review, irrigation planning and early investigation of unusual vegetation decline.

River-Basin Resilience

Palar Basin monitoring highlights riparian vegetation changes that can guide groundwater and riverbed conservation review.

Forest Monitoring

Jawadhu Foothills profiles help track seasonal canopy behaviour and identify periods that may need closer field inspection.

Wetland & Watershed Care

Otteri Lake monitoring adds NDVI and rainfall context for catchment runoff, tank storage and seasonal moisture planning.

Seasonal Climate Analysis

Monthly NDVI, temperature and rainfall profiles help interpret the ecosystem response across Vellore monsoon phases.

District Decision Support

Sector reports combine score summaries, time-series output, weather context and local recommendations for review.

Implementation — Current Backend Logic

CODE

1. Model Definition	`self.encoder = nn.LSTM(3, 64, batch_first=True)` `self.decoder = nn.LSTM(64, 3, batch_first=True)`	*The encoder compresses each five-step window to a 64-unit state. The decoder reconstructs the three input features.*

2. Fusion Score	`score = (`
	`  0.50 * reconstruction_mse`
	`  + 0.30 * z_score`
	`  + 0.20 * seasonal_deviation`
	`)`
		The active service fuses three signals. No Isolation Forest call appears in the current inference path.

3. Silence Score & Flag	`silence = MinMaxScaler().fit_transform(score) * 100`
	`threshold = mean(silence) + 1.5 * std(silence)`
	`anomaly = silence > threshold`
		The score is scaled per processed series. Health labels use <30, 30–70 and >70 bands.

Conclusion — Key Contributions

CHAPTER 5

01

Vellore-Specific Observatory

Fixed Vellore coordinates and four mapped ecosystems bring local river, forest, wetland and agricultural monitoring into one view across 213.5 km².

02

Three-Signal NDVI Fusion

A 64-unit LSTM Autoencoder combines reconstruction MSE, Z-score deviation and a 3-step rolling baseline into the 0–100 Silence Score.

03

Environmental & Geospatial Context

The dashboard combines 3D/2D sector mapping, Chart.js analytics and Open-Meteo telemetry with a 10-minute cache and offline fallback.

04

Working Full-Stack Prototype

FastAPI exposes NDVI inference, weather, map, CSV-upload and report endpoints; the frontend presents sector summaries and downloadable reports.

EchoSense turns local NDVI sequences into an interpretable monitoring workflow for Vellore ecosystems.

Future Work & Roadmap

ROADMAP

1. **Satellite NDVI Ingestion**

   Connect Sentinel-2 or Landsat-derived NDVI with timestamps, quality filtering and repeatable regional updates.

2. **Ground-Truth Validation**

   Build a labelled evaluation set from field observations so Precision, Recall, F1, ROC-AUC and false-alarm rates can be measured reliably.

3. **Model Comparison**

   Add Isolation Forest and other anomaly baselines as explicit comparators; evaluate them on the same labelled Vellore data.

4. **Persistent Data & Secure Access**

   Store sector histories and model outputs, and replace prototype login credentials with proper user management and secure authentication.

5. **Automated Alerts**

   Trigger email or webhook notifications for sustained high-risk scores and connect sector reports to local response workflows.